

Data Flow Diagrams (DFD)

Project: Offline-First, Proxy-Free College Attendance System

1. Level 0 DFD (Context Diagram)

Purpose: The Context Diagram provides a 10,000-foot view of the system. It shows the system as a single central process communicating with external entities (Student, Professor, Admin, and External APIs).

Entities and Data Flows:

- **Student:** Submits login credentials, hardware tokens (WebAuthn), and scanned attendance payloads. Receives sync status and attendance analytics.
- **Professor:** Submits class session requests and manual overrides. Receives the real-time dynamic QR stream, live headcount, and exported `.xlsx` files.
- **Admin:** Submits bulk rosters and approves device resets. Receives system audit logs.
- **Google Sheets (External):** Receives batched attendance rows.

Level 0 Visual Map (Mermaid)

```
graph LR
    %% Entities
    S[Student]
    P[Professor]
    A[Admin]
    G[Google Sheets API]

    %% Central Process
    SYS((0.0 <br> Proxy-Free <br> Attendance System))

    %% Flows from/to Student
    S -- "WebAuthn, Face Biometrics,<br>Scanned QR Payload" --> SYS
    SYS -- "Sync Status,<br>Overall Attendance %" --> S

    %% Flows from/to Professor
    P -- "Session Config,<br>Manual Overrides,<br>Export Cmd" --> SYS
    SYS -- "Dynamic QR Stream,<br>Live Headcount,<br>Excel (.xlsx)" --> P

    %% Flows from/to Admin
    A -- "Bulk CSV Rosters,<br>Device Reset Approvals" --> SYS
    SYS -- "Audit Logs,<br>Fraud Alerts" --> A

    %% Flows to Google Sheets
    SYS -- "Batched Verified Logs" --> G
```

2. Level 1 DFD (System Sub-Processes)

Purpose: The Level 1 DFD breaks down the central system into its primary sub-processes and introduces the Database Stores. This is what your professor will look at to understand how data moves internally.

Database Stores (D1-D4):

- **D1 - Users DB:** Stores profiles, passwords, WebAuthn public keys, and 128D face vectors.
- **D2 - Classes DB:** Stores course mappings and Professor/Student enrollment mapping.
- **D3 - Sessions DB:** Stores active/past class sessions, start times, and course metadata.
- **D4 - Attendance Logs DB:** Stores the actual verified check-ins and the `is_synced` status.

Main Processes (1.0 - 5.0):

- **1.0 Manage Auth & Device Binding:** Handles WebAuthn 1:1 hardware locking and initial logins.
- **2.0 Manage Lecture Sessions:** Validates Professor requests and generates the WebSocket dynamic QR token stream.
- **3.0 Process & Validate Attendance (Core Engine):** Validates the encrypted offline payloads, checks the timestamp/nonce, and enforces biometric matching.
- **4.0 Data Sync & Export:** Background worker that generates Excel files and communicates with Google Sheets.
- **5.0 Admin Operations:** Manages bulk uploads and device reset workflows.

Level 1 Visual Map (Mermaid)

```
graph LR
    %% --- Style Settings ---
    classDef actor fill:#f9f,stroke:#333,stroke-width:1px;
    classDef process fill:#bbf,stroke:#333,stroke-width:1px;
    classDef database fill:#dfd,stroke:#333,stroke-width:1px;

    %% --- Actors ---
    Student[Student]:::actor
    Prof[Professor]:::actor
    Admin[Admin]:::actor
    GSheets[Google Sheets API]:::actor

    %% --- SUBFLOW 1: Authentication & Admin ---
    subgraph Auth & Administration
        P1((1.0 Manage Auth<br>& Device Binding)):::process
        P5((5.0 Admin<br>Operations)):::process
        D1_A[(D1: Users DB)]:::database
        D2_A[(D2: Classes DB)]:::database

        Student -- "Login + WebAuthn Key" --> P1
        P1 -- "Auth Token" --> Student
        P1 == "Read/Write Keys" ==> D1_A

        Admin -- "Upload Bulk CSV,<br>Approve Resets" --> P5
        Student -- "Request Device Reset" --> P5
    end
```

```

P5 -- "Clear Device_Hash" --> D1_A
P5 -- "Update Rosters" --> D2_A
end

%% --- SUBFLOW 2: Lecture Sessions ---
subgraph Session Management
  P2((2.0 Manage<br>Lecture Sessions)):::process
  D2_B[(D2: Classes DB)]:::database
  D3_B[(D3: Sessions DB)]:::database

  Prof -- "Start Session<br>(Course ID)" --> P2
  P2 -- "Verify Assignment" --> D2_B
  P2 -- "Create Active Session" --> D3_B
  P2 -- "Stream 5s<br>Dynamic QR" --> Prof
end

%% --- SUBFLOW 3: Attendance Payload ---
subgraph Payload Validation
  P3((3.0 Validate<br>Attendance Payload)):::process
  D1_C[(D1: Users DB)]:::database
  D3_C[(D3: Sessions DB)]:::database
  D4_C[(D4: Attendance Logs DB)]:::database

  Student -- "Offline Payload<br>(QR Nonce, Time, Biometrics)" --> P3
  P3 -- "Validate Device ID" --> D1_C
  P3 -- "Validate Session & Time" --> D3_C
  P3 -- "Save Verified Log" --> D4_C
  P3 -- "Live Counter Update" --> Prof
end

%% --- SUBFLOW 4: Sync & Export ---
subgraph Sync & Report Export
  P4((4.0 Offline Sync<br>& Report Export)):::process
  D4_D[(D4: Attendance Logs DB)]:::database

  Prof -- "Request Export" --> P4
  P4 -- "Format .xlsx" --> Prof
  P4 == "Query & Mark<br>Unsynced Records" ==> D4_D
  P4 -- "Batch Append" --> GSheets
  GSheets -- "200 OK" --> P4
end

```